

Lessons 8–10. Partial Differential Equations I, II, and III

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1 Introduction to Partial Differential Equations

1.1 What Is a Partial Differential Equation?

So far in this course we have studied *ordinary* differential equations, where the unknown is a function of a single variable. We now make a decisive step forward and allow the unknown to depend on *two or more* independent variables. The corresponding equations are called **partial differential equations**, or PDEs.

Definition 1.1 (Partial Differential Equation). A **partial differential equation** of order k in the unknown $u = u(x_1, \dots, x_n)$ is an equation of the form

$$F\left(x_1, \dots, x_n, u, \frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_n}, \frac{\partial^2 u}{\partial x_1^2}, \dots, \frac{\partial^k u}{\partial x_n^k}\right) = 0,$$

where F is a given function and the order k is the highest order of any partial derivative appearing in the equation.

A **solution** of a PDE is any sufficiently differentiable function u that, when substituted into the equation, satisfies it identically on the domain of interest.

1.2 Classification of PDEs

Not all PDEs are alike, and recognising what type of equation you are dealing with is often the first step toward solving it. We distinguish several important classes.

Definition 1.2 (Linear, Quasilinear, Semilinear, and Nonlinear PDEs). Let the equation involve an unknown u and its partial derivatives.

- **Linear:** The equation is linear in u and in all its partial derivatives (with coefficients that may depend on the independent variables). That is, no products of u with its derivatives, and no nonlinear functions of u , appear.
- **Quasilinear:** The equation is linear in the highest-order derivatives of u , but the coefficients of those derivatives may depend on u itself and on lower-order derivatives.
- **Semilinear:** A quasilinear equation in which the coefficients of the highest-order derivatives depend only on the independent variables, not on u .
- **Nonlinear:** None of the above — the equation involves nonlinear functions of u or its derivatives at the highest order.

A linear PDE is called **homogeneous** if every term contains u or one of its derivatives; otherwise it is called **non-homogeneous** (or *inhomogeneous*). For example, $u_t + cu_x = 0$ is linear and homogeneous, while $u_t + cu_x = \sin(x)$ is linear and non-homogeneous.

1.3 Types of Solutions

Just as for ODEs, the *general solution* of a PDE involves arbitrary functions (not merely arbitrary constants), reflecting the infinite-dimensional freedom we have when solving a PDE. Imposing boundary or initial conditions pins down specific solutions.

- **General solution:** Expresses the solution in terms of arbitrary functions. For an n -th order PDE in two variables, the general solution typically involves n arbitrary functions of one variable.
- **Particular solution:** Obtained from the general solution by choosing specific forms for the arbitrary functions, usually to satisfy given initial or boundary conditions.
- **Singular solution:** A solution that cannot be obtained from the general solution by any choice of the arbitrary functions.

1.4 Two Introductory Examples

Let us make these ideas concrete with two elementary examples before we develop any systematic theory.

Example 1.1 (The Transport Equation). Consider the first-order linear PDE

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0,$$

where c is a positive constant. This is the **transport** (or **advection**) **equation**: it describes how a signal or quantity moves to the right at speed c without changing shape.

Solution. Let us guess that the solution has the form $u(x, t) = f(x - ct)$ for some differentiable function f . We verify by the chain rule:

$$\frac{\partial u}{\partial t} = f'(x - ct) \cdot (-c) = -cf'(x - ct), \quad \frac{\partial u}{\partial x} = f'(x - ct) \cdot 1 = f'(x - ct).$$

Therefore

$$u_t + c u_x = -cf'(x - ct) + cf'(x - ct) = 0. \quad \checkmark$$

The general solution is $u(x, t) = f(x - ct)$ for *any* differentiable function f . Given the initial condition $u(x, 0) = \phi(x)$, we immediately read off $f = \phi$, so

$$u(x, t) = \phi(x - ct).$$

The initial profile ϕ is transported to the right at speed c , unchanged in shape. This is our first encounter with the *method of characteristics*: the curves $x - ct = \text{const}$ are the **characteristics** of this PDE, and the solution is constant along each characteristic.

Example 1.2 (A Second-Order PDE Integrated Directly). Consider the second-order PDE

$$\frac{\partial^2 u}{\partial x \partial y} = 2x - y,$$

supplemented by the boundary conditions $u(x, 0) = 2 \sin(x)$ and $u(0, y) = 3y^4 - 5$.

Solution. We integrate the PDE with respect to y , holding x fixed:

$$\frac{\partial u}{\partial x} = 2xy - \frac{y^2}{2} + \phi(x),$$

where $\phi(x)$ is an arbitrary function arising from the integration. Integrating once more, now with respect to x :

$$u(x, y) = x^2 y - \frac{xy^2}{2} + \Phi(x) + G(y),$$

where $\Phi'(x) = \phi(x)$ is again an arbitrary function of x , and $G(y)$ is an arbitrary function of y . This is the general solution; we write it as

$$u(x, y) = x^2 y - \frac{xy^2}{2} + F(x) + G(y).$$

Now we apply the boundary conditions to determine F and G .

From $u(x, 0) = 2 \sin(x)$:

$$x^2 \cdot 0 - x \cdot 0 + F(x) + G(0) = 2 \sin(x).$$

Taking $G(0) = 0$, we obtain $F(x) = 2 \sin(x)$.

From $u(0, y) = 3y^4 - 5$:

$$0 - 0 + F(0) + G(y) = 3y^4 - 5.$$

Since $F(0) = 2 \sin(0) = 0$, we get $G(y) = 3y^4 - 5$.

The particular solution satisfying both boundary conditions is therefore

$$u(x, y) = x^2 y - \frac{xy^2}{2} + 2 \sin(x) + 3y^4 - 5.$$

2 First-Order PDEs: Classification

2.1 The Standard Form

A general first-order PDE in two independent variables x and y can be written as

$$F(x, y, u, u_x, u_y) = 0.$$

The most practically important subclass is the **quasilinear** first-order PDE, which in standard form reads

$$a(x, y, u) u_x + b(x, y, u) u_y = c(x, y, u). \quad (1)$$

The equation is quasilinear because the coefficients a , b , c may depend on the unknown u , but the equation is linear in the first-order derivatives u_x and u_y .

2.2 A Hierarchy of Special Cases

Within the family (1), there is a natural hierarchy depending on how the coefficients depend on u :

1. **Linear with constant coefficients:** a , b are constants, and $c = c_1 u + c_2$ with c_1 , c_2 constants. Example: $3u_x - 2u_y = 7u$.
2. **Linear with variable coefficients:** $a = a(x, y)$, $b = b(x, y)$, and $c = c_1(x, y) u + c_2(x, y)$. Example: $x u_x + y^2 u_y = (x + y)u + \cos x$.
3. **Semilinear:** $a = a(x, y)$, $b = b(x, y)$, but $c = c(x, y, u)$ may depend nonlinearly on u . The key point is that the principal part $a u_x + b u_y$ is linear. Example: $u_x + u_y = u^2$.
4. **Quasilinear:** The general form (1) with a , b , c all possibly depending on u . Example: $u u_x + u_y = 1$ (the characteristics now depend on the solution itself).
5. **Nonlinear (fully):** The PDE is genuinely nonlinear in the first-order derivatives. Example: $(u_x)^2 + (u_y)^2 = 1$ (the eikonal equation).

2.3 The Inviscid Burgers' Equation

One of the most celebrated quasilinear first-order PDEs is the **inviscid Burgers' equation**:

$$u_t + u u_x = 0.$$

Here t plays the role of time and x is a spatial coordinate. The coefficient of u_x is the solution u itself, making this equation genuinely quasilinear. Despite its apparent simplicity, it captures the essential mechanism of *wave steepening* and *shock formation*: a smooth initial condition can develop a vertical slope in finite time, modelling phenomena such as traffic jams or the steepening of ocean waves. We will solve an initial value problem for Burgers' equation in the next section using the method of characteristics.

3 The Method of Characteristics

3.1 Motivation and Geometric Picture

Let us think geometrically. A solution $u = u(x, y)$ of (1) can be regarded as a surface $\mathcal{S}$ in (x, y, u) -space: the *solution surface* or *integral surface*. The normal to $\mathcal{S}$ at any point (x_0, y_0, u_0) is proportional to $(u_x, u_y, -1)$. The PDE (1) says that the vector $\mathbf{F} = (a, b, c)$ is *perpendicular* to this normal, i.e., $\mathbf{F}$ lies tangent to the surface. Since $\mathbf{F}$ is tangent to $\mathcal{S}$ at every point, the integral curves of the vector field $\mathbf{F}$ lie entirely within $\mathcal{S}$. These integral curves are the **characteristic curves** of the PDE.

Following the tangent vector $\mathbf{F} = (a, b, c)$ along the solution surface gives the system

$$\frac{dx}{ds} = a(x, y, u), \quad \frac{dy}{ds} = b(x, y, u), \quad \frac{du}{ds} = c(x, y, u),$$

or equivalently in the non-parametric ratio form,

$$\frac{dx}{a} = \frac{dy}{b} = \frac{du}{c}. \quad (2)$$

These are the **characteristic equations** of the PDE (1).

3.2 Semi-Linear Case: Finding the General Solution

For a semi-linear PDE $a(x, y) u_x + b(x, y) u_y = c(x, y, u)$, the characteristic equations (2) split into two steps:

1. Solve $dx/a = dy/b$ for a first integral $\varphi(x, y) = C_1$.
2. Solve $dy/b = du/c$ (substituting the expression for the characteristic in step 1) for a second first integral $\psi(x, y, u) = C_2$.

The general solution is then $C_2 = F(C_1)$, i.e., $\psi(x, y, u) = F(\varphi(x, y))$, for an arbitrary function F . Applying an initial or boundary condition determines F uniquely.

Example 3.1 (Semi-Linear PDE with Initial Condition). Solve $2u_x + 3u_y = u$ subject to $u(x, 0) = f(x)$.

Step 1 — Characteristic equations. Here $a = 2$, $b = 3$, $c = u$, so

$$\frac{dx}{2} = \frac{dy}{3} = \frac{du}{u}.$$

Step 2 — First pair. From $dx/2 = dy/3$:

$$3 dx = 2 dy \implies 3x - 2y = C_1.$$

Step 3 — Second pair. From $dy/3 = du/u$:

$$\frac{du}{u} = \frac{dy}{3} \implies \ln |u| = \frac{y}{3} + \ln |C_2| \implies u = C_2 e^{y/3}.$$

Step 4 — General solution. Writing $C_2 = F(C_1)$:

$$u(x, y) = e^{y/3} F(3x - 2y).$$

Step 5 — Apply initial condition. Setting $y = 0$:

$$u(x, 0) = e^0 F(3x) = f(x) \implies F(3x) = f(x) \implies F(\xi) = f\left(\frac{\xi}{3}\right).$$

Therefore

$$u(x, y) = e^{y/3} f\left(\frac{3x - 2y}{3}\right) = e^{y/3} f\left(x - \frac{2y}{3}\right).$$

One can verify that this formula encodes exactly the transport observed earlier: the solution profile f is advected along the lines $x - 2y/3 = \text{const}$, while being amplified exponentially as y grows.

3.3 Quasi-Linear Case: Parametric Form

For the fully quasi-linear equation, the coefficients a, b, c depend on u , so the characteristic curves in (x, y, u) -space are curves, not just families parameterised by an independent integration. It is convenient to use a *parametric* formulation. Let the initial curve $\mathcal{C}$ in (x, y) -space be given parametrically by $(x_0(\xi), y_0(\xi))$, with the initial value $u_0(\xi)$ prescribed there. We seek characteristic curves $(X(s, \xi), Y(s, \xi), Z(s, \xi))$ in (x, y, u) -space satisfying

$$\frac{dX}{ds} = a(X, Y, Z), \quad \frac{dY}{ds} = b(X, Y, Z), \quad \frac{dZ}{ds} = c(X, Y, Z), \quad (3)$$

with initial conditions $X(0, \xi) = x_0(\xi)$, $Y(0, \xi) = y_0(\xi)$, $Z(0, \xi) = u_0(\xi)$.

Solving (3) gives the characteristic curves. By the implicit function theorem (assuming the Jacobian $\partial(X, Y)/\partial(s, \xi) \neq 0$), we can invert $s = S(x, y)$, $\xi = \Xi(x, y)$ and express $u = Z(S(x, y), \Xi(x, y))$.

Definition 3.1 (Projected Characteristics). The projection of a characteristic curve $(X(s), Y(s), Z(s))$ onto the (x, y) -plane (i.e., ignoring the Z -component) is called a **projected characteristic curve**, or simply a **characteristic**.

Example 3.2 (Quasi-Linear Equation — General Solution). Find the general solution to $2y u_x + u u_y = 2y u^2$.

Identification. Here $a = 2y$, $b = u$, $c = 2y u^2$. The characteristic equations are

$$\frac{dx}{2y} = \frac{dy}{u} = \frac{du}{2y u^2}.$$

Second pair. We use $dy/u = du/(2yu^2)$:

$$\frac{du}{dy} = \frac{2yu^2}{u} = 2yu \implies \frac{du}{u} = 2y dy \implies \ln|u| = y^2 + \ln|C_2| \implies u = C_2 e^{y^2}.$$

First pair. Now substitute $u = C_2 e^{y^2}$ into $dx/2y = dy/u$:

$$dx = \frac{2y}{u} dy = \frac{2y}{C_2 e^{y^2}} dy = \frac{2y e^{-y^2}}{C_2} dy.$$

Integrating both sides: $C_2 x = -e^{-y^2} + C_1$, so

$$C_1 = C_2 x + e^{-y^2} = u e^{-y^2} \cdot x + e^{-y^2} = e^{-y^2} (xu + 1).$$

General solution. Writing $C_2 = F(C_1)$ and using $C_2 = u e^{-y^2}$:

$$u e^{-y^2} = F(e^{-y^2} (1 + xu)),$$

or equivalently,

$$u(x, y) = e^{y^2} F((1 + xu) e^{-y^2}),$$

where F is an arbitrary function. This is an *implicit* relation for u ; the function F would be determined by an initial or boundary condition.

Remark 3.1. Every one-parameter family of characteristic curves generates an integral surface of the PDE. Conversely, every solution surface may be viewed as such a family. This is the deep geometric content of the method of characteristics.

Example 3.3 (Burgers' Equation: An Initial Value Problem). Solve the inviscid Burgers' equation

$$u_t + u u_x = 0, \quad u(x, 0) = 1 - x^2.$$

Step 1 — Parametric characteristic equations. Using t in place of y , and treating u as the Z -component, the system (3) becomes

$$\frac{dT}{ds} = 1, \quad \frac{dX}{ds} = Z, \quad \frac{dZ}{ds} = 0,$$

with initial conditions $T(0) = 0$, $X(0) = \xi$, $Z(0) = 1 - \xi^2$ (here ξ parametrises the initial curve, i.e., the x -axis).

Step 2 — Solve.

$$T(s) = s, \quad Z(s) = 1 - \xi^2, \quad X(s) = (1 - \xi^2) s + \xi.$$

In terms of (t, x) : $s = t$ and

$$x = (1 - \xi^2) t + \xi.$$

Step 3 — Invert for ξ . Rearranging:

$$t\xi^2 - \xi + (x - t) = 0.$$

The quadratic formula gives

$$\xi = \frac{1 \pm \sqrt{1 - 4t(x - t)}}{2t} = \frac{1 \pm \sqrt{1 + 4t^2 - 4tx}}{2t}.$$

Step 4 — Express u . Since $u = Z = 1 - \xi^2$:

$$u(t, x) = 1 - \left(\frac{1 \pm \sqrt{1 + 4t^2 - 4tx}}{2t} \right)^2 = \frac{2tx - 1 \mp \sqrt{1 + 4t^2 - 4tx}}{2t^2}.$$

The physically relevant branch (continuous with the initial data for small $t > 0$) is selected by requiring $u \rightarrow 1 - x^2$ as $t \rightarrow 0^+$.

Remark 3.2. This solution is valid only as long as the characteristics do not cross. When characteristics emanating from different initial points ξ reach the same (t, x) , the gradient u_x blows up and a **shock** forms. The formula then ceases to be a classical solution.

4 Classification of Second-Order PDEs

4.1 The General Second-Order Linear PDE

We now turn to second-order PDEs, which form the heart of classical mathematical physics. A general second-order linear PDE in two independent variables x and y has the form

$$a u_{xx} + 2b u_{xy} + c u_{yy} + d u_x + e u_y + f u = g, \quad (4)$$

where a, b, c, d, e, f, g may be functions of x and y (but not of u). The *principal part* is $au_{xx} + 2bu_{xy} + cu_{yy}$, and the classification depends entirely on this part.

4.2 The Discriminant and Three Types

Definition 4.1 (Classification by Discriminant). The discriminant of equation (4) at a given point (x_0, y_0) is

$$D = ac - b^2.$$

The PDE is classified as:

- **Hyperbolic** at (x_0, y_0) if $D < 0$,
- **Parabolic** at (x_0, y_0) if $D = 0$,
- **Elliptic** at (x_0, y_0) if $D > 0$.

Let us connect this abstract classification to the three canonical equations of mathematical physics.

- **Wave equation** $u_{tt} = c^2 u_{xx}$: Here $a = 1$, $b = 0$, $c = -c^2$ (careful with notation — the constant c in the equation and the coefficient c in the formula are different things; we write $D = 1 \cdot (-c^2) - 0^2 = -c^2 < 0$). So the wave equation is hyperbolic. Hyperbolic equations describe wave propagation and have a finite speed of information travel.
- **Heat equation** $u_t = k u_{xx}$: This can be written as $-k u_{xx} + u_t = 0$. With x and t as the two variables, the principal part has $a = -k$, $b = 0$, $c = 0$ (coefficient of u_{tt} is zero!), giving $D = 0$. Hence the heat equation is parabolic. Parabolic equations describe diffusion; information travels at infinite speed but with decreasing influence over distance.

- **Laplace's equation** $u_{xx} + u_{yy} = 0$: Here $a = c = 1$, $b = 0$, so $D = 1 > 0$. Laplace's equation is elliptic. Elliptic equations describe steady states; there are no propagating waves, and the solution is smooth everywhere in the interior (assuming smooth data).

Example 4.1 (A PDE of Mixed Type). Classify the PDE $u_{xx} + y u_{yy} + \frac{1}{2}u_y = 0$.

Solution. The principal part coefficients are $a = 1$, $b = 0$, $c = y$. The discriminant is $D = ac - b^2 = 1 \cdot y - 0 = y$.

- If $y > 0$: $D > 0 \Rightarrow$ **elliptic**.
- If $y = 0$: $D = 0 \Rightarrow$ **parabolic**.
- If $y < 0$: $D < 0 \Rightarrow$ **hyperbolic**.

This is a *PDE of mixed type*: it changes character depending on the region of the (x, y) -plane. Such equations arise in transonic aerodynamics (the Tricomi equation is the prototype).

5 Initial and Boundary Conditions

5.1 Why Conditions Are Needed

A PDE alone admits infinitely many solutions — the general solution contains arbitrary functions rather than constants. To single out a physically meaningful unique solution, we must impose additional *data* on part of the boundary of the domain. The type of data appropriate for a PDE depends crucially on whether the equation is hyperbolic, parabolic, or elliptic.

5.2 Types of Boundary Conditions

Consider a PDE defined on a domain Ω with boundary $\partial\Omega$.

Definition 5.1 (Boundary Condition Types). • **Dirichlet condition**: the value of u is prescribed on (part of) the boundary,

$$u|_{\partial\Omega} = g(x, y).$$

- **Neumann condition**: the normal derivative $\partial u / \partial n$ is prescribed,

$$\left. \frac{\partial u}{\partial n} \right|_{\partial\Omega} = h(x, y).$$

- **Robin (mixed) condition**: a linear combination is prescribed,

$$\alpha u + \beta \left. \frac{\partial u}{\partial n} \right|_{\partial\Omega} = \gamma(x, y), \quad \alpha^2 + \beta^2 > 0.$$

5.3 Initial, Boundary, and Initial-Boundary Value Problems

- An **initial value problem (IVP)** or **Cauchy problem** specifies u (and possibly u_t) at some initial time $t = 0$ over the whole spatial domain. This is the natural setting for hyperbolic and parabolic equations.
- A **boundary value problem (BVP)** specifies u or its derivatives on the boundary of a spatial domain, with no time evolution. This is the natural setting for elliptic equations.
- An **initial-boundary value problem (IBVP)** combines both: initial data at $t = 0$ together with boundary conditions for all $t > 0$ on a bounded spatial domain.

Definition 5.2 (Well-Posedness (Hadamard)). A problem for a PDE is called **well-posed** if:

1. A solution **exists**.
2. The solution is **unique**.
3. The solution depends **continuously** on the data (stability).

If any of these three conditions fails, the problem is called **ill-posed**. Much of the theory of PDEs is devoted to proving well-posedness for physically meaningful problems.

6 The One-Dimensional Heat Equation

6.1 Physical Derivation

Imagine a thin, homogeneous wire of length L lying along the x -axis, $0 \leq x \leq L$. The wire is insulated along its lateral surface, so heat flows only in the x -direction. Let $u(x, t)$ denote the temperature at position x and time t . From Fourier's law of heat conduction and conservation of energy, one derives the **heat equation** (also called the **diffusion equation**):

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < L, \quad t > 0,$$

where $k > 0$ is the *thermal diffusivity* of the material.

6.2 IBVP with Zero Dirichlet Boundary Conditions

The simplest situation is when the ends of the wire are kept at temperature zero. The full IBVP reads:

$$u_t = k u_{xx}, \quad 0 < x < L, \quad t > 0, \tag{5}$$

$$u(0, t) = 0, \quad u(L, t) = 0, \quad t > 0, \tag{6}$$

$$u(x, 0) = f(x), \quad 0 < x < L. \tag{7}$$

6.2.1 Separation of Variables

We look for solutions of the form $u(x, t) = X(x) T(t)$. Substituting into the PDE:

$$X(x) T'(t) = k X''(x) T(t) \implies \frac{T'(t)}{k T(t)} = \frac{X''(x)}{X(x)} = -\lambda^2,$$

where $-\lambda^2$ is the separation constant (we choose the sign to obtain oscillatory solutions in x consistent with the boundary conditions). This yields two ODEs:

$$X''(x) + \lambda^2 X(x) = 0, \quad T'(t) + \lambda^2 k T(t) = 0.$$

6.2.2 Eigenvalue Problem for X

The boundary conditions $u(0, t) = u(L, t) = 0$ require $X(0) = X(L) = 0$.

- **Case** $\lambda^2 < 0$: $X = C_1 e^{\mu x} + C_2 e^{-\mu x}$ with $\mu > 0$. The boundary conditions force $C_1 = C_2 = 0$: only the trivial solution.
- **Case** $\lambda^2 = 0$: $X = C_1 x + C_2$. Then $X(0) = 0 \Rightarrow C_2 = 0$ and $X(L) = 0 \Rightarrow C_1 = 0$: again trivial.
- **Case** $\lambda^2 > 0$: $X = C_1 \cos(\lambda x) + C_2 \sin(\lambda x)$. From $X(0) = 0$: $C_1 = 0$. From $X(L) = 0$: $C_2 \sin(\lambda L) = 0$. For a non-trivial solution, $\sin(\lambda L) = 0$, so

$$\lambda = \lambda_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

and the corresponding eigenfunctions are

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right).$$

6.2.3 Solving for T_n

For each $\lambda_n = n\pi/L$, the equation for T becomes

$$T'_n(t) + \frac{n^2 \pi^2 k}{L^2} T_n(t) = 0 \implies T_n(t) = a_n e^{-n^2 \pi^2 k t / L^2}.$$

6.2.4 The Full Solution

By the superposition principle, the general solution (a series of product solutions) is

$$u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right) e^{-n^2 \pi^2 k t / L^2}.$$

Setting $t = 0$ and requiring $u(x, 0) = f(x)$:

$$f(x) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right).$$

This is the **Fourier sine series** of f on $[0, L]$. The coefficients are given by

$$a_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx, \quad n = 1, 2, 3, \dots$$

Theorem 6.1 (Solution of the Dirichlet Heat Problem). *The solution of the IBVP (5)–(7) is*

$$u(x, t) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right) e^{-n^2\pi^2 k t/L^2}, \quad (8)$$

where the Fourier sine coefficients are

$$a_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

Remark 6.1. The exponential factors $e^{-n^2\pi^2 k t/L^2}$ decay *super-polynomially* in n for any $t > 0$. This means that even if f is merely piecewise continuous (and its Fourier series converges only conditionally), the series (8) converges absolutely and defines a *smooth* function $u(\cdot, t) \in C^\infty(0, L)$ for every $t > 0$. The heat equation thus has an instantaneous **smoothing effect**: all irregularities in the initial data are immediately ironed out.

6.3 IBVP with Zero Neumann Boundary Conditions (Insulated Ends)

Now suppose both ends of the wire are perfectly insulated (no heat flux through the endpoints). The boundary conditions become:

$$u_x(0, t) = 0 \quad \text{and} \quad u_x(L, t) = 0, \quad t > 0.$$

Separating variables gives the same ODEs, but the eigenvalue problem for X now reads: $X'' + \lambda^2 X = 0$, $X'(0) = X'(L) = 0$.

- **Case $\lambda^2 = 0$:** $X = C_1 x + C_2$, so $X' = C_1$. The conditions $X'(0) = X'(L) = 0$ give $C_1 = 0$, leaving $X_0 = 1$ (a nontrivial constant solution). The corresponding $T_0(t) = a_0$ is a constant.
- **Case $\lambda^2 > 0$:** $X = C_1 \cos(\lambda x) + C_2 \sin(\lambda x)$; then $X' = -C_1 \lambda \sin(\lambda x) + C_2 \lambda \cos(\lambda x)$. From $X'(0) = C_2 \lambda = 0$: $C_2 = 0$. From $X'(L) = -C_1 \lambda \sin(\lambda L) = 0$: $\sin(\lambda L) = 0$, giving $\lambda_n = n\pi/L$ and

$$X_n(x) = \cos\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, \dots$$

Theorem 6.2 (Solution of the Neumann Heat Problem). *The solution of the heat IBVP with insulated-end boundary conditions is*

$$u(x, t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) e^{-n^2\pi^2 k t/L^2}, \quad (9)$$

where the Fourier cosine coefficients are

$$a_0 = \frac{1}{L} \int_0^L f(x) dx, \quad a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx.$$

Notice that the $n = 0$ term a_0 is the spatial average of the initial temperature. As $t \rightarrow \infty$, all exponentials decay to zero and $u(x, t) \rightarrow a_0$: the temperature equilibrates to the spatial mean, as we expect physically when the wire is perfectly insulated (total heat is conserved).

Example 6.1 (Insulated Wire). Consider a wire of length $L = 1$ with diffusivity $k = 0.25$ and insulated ends. The initial temperature distribution is $f(x) = 100x(1 - x)$. Find $u(x, t)$.

Solution. We compute the cosine coefficients.

Constant term:

$$a_0 = \int_0^1 100x(1 - x) dx = 100 \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 100 \left(\frac{1}{2} - \frac{1}{3} \right) = 100 \cdot \frac{1}{6} = \frac{50}{3}.$$

Cosine coefficients ($n \geq 1$): Using integration by parts twice (or consulting a table of Fourier integrals):

$$a_n = 2 \int_0^1 100x(1 - x) \cos(n\pi x) dx = -\frac{200(1 + (-1)^n)}{n^2\pi^2}.$$

Observe that $a_n = 0$ when n is odd, and $a_n = -400/(n^2\pi^2)$ when n is even. Physically, the initial profile is symmetric about $x = 1/2$, so only even cosine modes (which are also symmetric) are excited.

Full solution:

$$u(x, t) = \frac{50}{3} - \frac{200}{\pi^2} \sum_{n=1}^{\infty} \frac{1 + (-1)^n}{n^2} e^{-n^2\pi^2 \cdot 0.25t} \cos(n\pi x).$$

As $t \rightarrow \infty$, $u(x, t) \rightarrow 50/3 \approx 16.7$, the uniform equilibrium temperature.

7 The One-Dimensional Wave Equation

7.1 Physical Derivation: The Vibrating String

Consider a flexible elastic string of length L , stretched under constant tension τ and fixed at both endpoints. Let $\rho(x)$ denote its mass density, and $u(x, t)$ its transverse displacement. Applying Newton's second law to a small segment of the string and taking the limit as the segment length goes to zero, one derives

$$\rho(x) \frac{\partial^2 u}{\partial t^2} = \tau \frac{\partial^2 u}{\partial x^2}.$$

For a uniform string (ρ constant), we set $c^2 = \tau/\rho$ and obtain the **wave equation**:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

The constant $c = \sqrt{\tau/\rho}$ is the **wave speed**: the speed at which disturbances travel along the string. The wave equation is the prototype hyperbolic PDE.

7.2 The IBVP

The complete problem for a string of length L fixed at both ends, with prescribed initial displacement $f(x)$ and initial velocity $g(x)$, reads:

$$u_{tt} = c^2 u_{xx}, \quad 0 < x < L, \quad t > 0, \quad (10)$$

$$u(0, t) = 0, \quad u(L, t) = 0, \quad t > 0, \quad (11)$$

$$u(x, 0) = f(x), \quad u_t(x, 0) = g(x), \quad 0 < x < L. \quad (12)$$

7.3 Separation of Variables

We try $u(x, t) = X(x)T(t)$. Substituting:

$$\frac{T''(t)}{c^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda^2.$$

This gives

$$X'' + \lambda^2 X = 0, \quad T'' + \lambda^2 c^2 T = 0.$$

The eigenvalue problem $X'' + \lambda^2 X = 0$, $X(0) = X(L) = 0$ is identical to the heat equation case. The eigenvalues and eigenfunctions are

$$\lambda_n = \frac{n\pi}{L}, \quad X_n(x) = \sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, 3, \dots$$

For each λ_n , the equation for T is $T'' + \lambda_n^2 c^2 T = 0$, which is simple harmonic motion with angular frequency $\omega_n = \lambda_n c = n\pi c/L$. The general solution is

$$T_n(t) = b_n \cos\left(\frac{n\pi c t}{L}\right) + b_n^* \sin\left(\frac{n\pi c t}{L}\right).$$

7.3.1 The Full Solution

By superposition:

$$u(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) \left[b_n \cos\left(\frac{n\pi c t}{L}\right) + b_n^* \sin\left(\frac{n\pi c t}{L}\right) \right]. \quad (13)$$

Applying the initial conditions:

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right),$$

$$u_t(x, 0) = g(x) = \sum_{n=1}^{\infty} b_n^* \cdot \frac{n\pi c}{L} \cdot \sin\left(\frac{n\pi x}{L}\right).$$

Reading off the Fourier sine coefficients:

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx, \quad b_n^* = \frac{2}{n\pi c} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

Theorem 7.1 (Solution of the Wave IBVP). *The solution of (10)–(12) is given by (13), where*

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx, \quad b_n^* = \frac{2}{n\pi c} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

Each term $X_n T_n$ is called the n -th **normal mode** or **standing wave**: the string oscillates sinusoidally in space with n half-wavelengths across $[0, L]$ and vibrates in time at frequency $f_n = nc/(2L)$. The first mode ($n = 1$) is the **fundamental frequency**; higher modes are **overtones**.

Example 7.1 (The Plucked String). A string of length $L = 2$ and wave speed $c = 1$ is plucked into the triangular shape

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ 2 - x, & 1 \leq x \leq 2, \end{cases}$$

and released from rest ($g(x) = 0$). Find $u(x, t)$.

Solution. Since $g = 0$, all $b_n^* = 0$. We compute b_n ; with $L = 2$:

$$b_n = \frac{2}{2} \int_0^2 f(x) \sin\left(\frac{n\pi x}{2}\right) dx = \int_0^1 x \sin\left(\frac{n\pi x}{2}\right) dx + \int_1^2 (2-x) \sin\left(\frac{n\pi x}{2}\right) dx.$$

Using integration by parts on each piece (the calculation is straightforward but careful), one obtains

$$b_n = \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right).$$

The sequence of values is $b_1 = 8/\pi^2$, $b_2 = 0$, $b_3 = -8/(9\pi^2)$, $b_4 = 0$, $\dots$ (only the odd modes are excited, reflecting the symmetry of the initial profile about $x = 1$). The solution is

$$u(x, t) = \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi x}{2}\right) \cos\left(\frac{n\pi t}{2}\right).$$

8 d'Alembert's Solution in an Infinite Domain

8.1 The General Solution on the Real Line

When there are no boundary conditions (the string is infinitely long, or we are looking at a problem where boundaries are far away), the wave equation $u_{tt} = c^2 u_{xx}$ on $\mathbb{R}$ can be solved completely and explicitly.

The key observation is that the wave operator factors:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = \left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x}\right) \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x}\right) u = 0.$$

Setting $\xi = x - ct$ and $\eta = x + ct$ as new coordinates (*characteristic coordinates*), the equation reduces to $\partial^2 u / \partial \xi \partial \eta = 0$, which integrates directly to

$$u(x, t) = F(x - ct) + G(x + ct),$$

for arbitrary functions F and G . The function $F(x - ct)$ represents a wave profile travelling to the right at speed c ; $G(x + ct)$ travels to the left. This is the **general solution** of the wave equation on $\mathbb{R}$.

8.2 d'Alembert's Formula

Given initial conditions $u(x, 0) = f(x)$ and $u_t(x, 0) = g(x)$, we can determine F and G explicitly.

At $t = 0$: $F(x) + G(x) = f(x)$. Differentiating with respect to t and evaluating at $t = 0$: $-cF'(x) + cG'(x) = g(x)$, so $G'(x) - F'(x) = g(x)/c$.

Integrating: $G(x) - F(x) = \frac{1}{c} \int_{x_0}^x g(s) ds + K$ for some constant K . Combining with $F + G = f$:

$$G(x) = \frac{f(x)}{2} + \frac{1}{2c} \int_{x_0}^x g(s) ds + \frac{K}{2}, \quad F(x) = \frac{f(x)}{2} - \frac{1}{2c} \int_{x_0}^x g(s) ds - \frac{K}{2}.$$

Substituting into $u = F(x - ct) + G(x + ct)$ and simplifying:

Theorem 8.1 (d'Alembert's Formula). *The solution of the Cauchy problem for the wave equation on $\mathbb{R}$,*

$$u_{tt} = c^2 u_{xx}, \quad u(x, 0) = f(x), \quad u_t(x, 0) = g(x),$$

is given by d'Alembert's formula:

$$u(x, t) = \frac{f(x - ct) + f(x + ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds.$$

Remark 8.1. The value $u(x, t)$ depends only on the initial data in the interval $[x - ct, x + ct]$. This interval is the **domain of dependence** of the point (x, t) . Conversely, the initial data at a point x_0 influences the solution only in the region $|x - x_0| \leq ct$, which is the **domain of influence**. This reflects the finite propagation speed c of the wave equation: signals cannot travel faster than c .

Example 8.1 (Gaussian Initial Displacement). Solve the Cauchy problem with $f(x) = e^{-x^2}$ (a Gaussian pulse) and $g(x) = 0$.

Solution. By d'Alembert's formula with $g = 0$:

$$u(x, t) = \frac{e^{-(x-ct)^2} + e^{-(x+ct)^2}}{2}.$$

At $t = 0$, this is simply e^{-x^2} . For $t > 0$, the Gaussian splits into two identical pulses of half the original height: one travelling to the right at speed c , centred at $x = ct$, and one to the left, centred at $x = -ct$. The two pulses separate and never interact again (on $\mathbb{R}$).

9 Smoothness: Heat Equation versus Wave Equation

9.1 A Tale of Two Equations

The heat and wave equations both arise as separation-of-variables series, yet they behave radically differently with regard to the smoothness of their solutions. Understanding this contrast is one of the most instructive lessons in the subject.

Let us compare the two on the interval $[0, \pi]$ with Dirichlet boundary conditions, using the *same* discontinuous initial data

$$f(x) = \begin{cases} x, & 0 \leq x < \pi/2, \\ 0, & \pi/2 \leq x \leq \pi. \end{cases}$$

The Fourier sine coefficients are $b_n = \frac{2}{\pi} \int_0^\pi f(x) \sin(nx) dx$, which we can compute (by parts) and which are of order $O(1/n)$ for large n , reflecting the jump discontinuity of f at $x = \pi/2$.

9.2 The Heat Equation Smooths Instantly

For the heat equation $u_t = u_{xx}$ (with $k = 1$ for simplicity) on $[0, \pi]$ with the same initial data:

$$u_{\text{heat}}(x, t) = \sum_{n=1}^{\infty} b_n e^{-n^2 t} \sin(nx).$$

For any $t > 0$, the coefficients $b_n e^{-n^2 t}$ decay faster than any power of $1/n$ (because $e^{-n^2 t}$ decays super-exponentially in n). By standard theorems on Fourier series, this means the series can be differentiated term-by-term any number of times, and $u(\cdot, t)$ is in $C^\infty(0, \pi)$ for every $t > 0$.

In plain terms: *no matter how rough the initial temperature distribution, the heat equation smooths it out instantly.* A delta function at $t = 0$ becomes a smooth Gaussian for any $t > 0$.

9.3 The Wave Equation Preserves Singularities

For the wave equation $u_{tt} = u_{xx}$ on $[0, \pi]$:

$$u_{\text{wave}}(x, t) = \sum_{n=1}^{\infty} b_n \cos(nt) \sin(nx).$$

The coefficients $b_n \cos(nt)$ have the *same* magnitude as b_n for all t : the cosines do not decay. Therefore the sum is no smoother than the original Fourier series of f . If f had a jump discontinuity at $x = \pi/2$, then $u(\cdot, t)$ will have jump discontinuities (at the “images” of $x = \pi/2$ along the characteristics $x \pm t = \text{const}$) for all time. Singularities travel along the characteristic lines $x = \pi/2 \pm t$ without dissipating.

This is consistent with d’Alembert’s formula: $u(x, t) = [f(x - t) + f(x + t)]/2$ (for $g = 0$), which inherits exactly the discontinuities of f .

9.4 Summary

Property	Heat equation	Wave equation
Type	Parabolic	Hyperbolic
Propagation speed	Infinite	Finite ($= c$)
Smoothness for $t > 0$	C^∞ (instant smoothing)	Same as initial data
Energy	Decreasing (dissipative)	Conserved
Long-time behaviour	$u \rightarrow 0$ (Dirichlet BCs)	Oscillatory forever

10 Laplace’s Equation

10.1 From Heat to Steady State

We have seen that the heat equation $u_t = k \Delta u$ describes how temperature evolves over time. A natural question is: what does the temperature look like in the long run, after it has had time to equilibrate? Mathematically, we seek a solution that does not depend on time, i.e., $u_t = 0$. Setting the time derivative to zero in the heat equation leaves:

$$\Delta u = 0,$$

the **Laplace equation**. Its solutions are called **harmonic functions**. Laplace’s equation is the canonical example of an elliptic PDE.

10.2 One-Dimensional Steady State

In one dimension, $\Delta u = u_{xx} = 0$ has the general solution

$$u(x) = C_1 x + C_2,$$

i.e., a straight line. For a wire with fixed endpoint temperatures T_1 at $x = 0$ and T_2 at $x = L$, the steady state is

$$u(x) = \frac{T_2 - T_1}{L} x + T_1.$$

This linear interpolation is completely expected: heat flows uniformly from the hotter end to the cooler end.

10.3 Laplace's Equation in Two Dimensions

In two spatial dimensions, the heat equation for a thin plate is

$$u_t = k(u_{xx} + u_{yy}) = k \Delta u,$$

where $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2$ is the **Laplacian**. The steady state satisfies

$$\Delta u = u_{xx} + u_{yy} = 0. \quad (14)$$

Definition 10.1 (Harmonic Function). A function $u = u(x, y)$ that satisfies $\Delta u = 0$ on an open domain Ω is called **harmonic** on Ω .

Harmonic functions are remarkable: they are infinitely differentiable wherever they satisfy the equation (elliptic regularity), they satisfy the mean-value property (the value at any point equals the average over any circle centred there), and they attain their maximum and minimum only on the boundary.

10.4 Dirichlet Problem in a Rectangle: Special Case

Let us solve Laplace's equation on the rectangle $[0, a] \times [0, b]$ where the temperature is zero on three sides and prescribed on the top edge:

$$u_{xx} + u_{yy} = 0, \quad u(x, 0) = 0, \quad u(x, b) = f_2(x), \quad u(0, y) = 0, \quad u(a, y) = 0. \quad (15)$$

10.4.1 Separation of Variables

Try $u(x, y) = X(x)Y(y)$. Substituting into (15)₁:

$$\frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = k.$$

The boundary conditions $u(0, y) = u(a, y) = 0$ require $X(0) = X(a) = 0$, and $u(x, 0) = 0$ requires $Y(0) = 0$.

For the X -equation $X'' + kX = 0$ with $X(0) = X(a) = 0$: only $k = \mu_n^2 > 0$ gives non-trivial solutions. Just as before,

$$\mu_n = \frac{n\pi}{a}, \quad X_n(x) = \sin\left(\frac{n\pi x}{a}\right), \quad n = 1, 2, \dots$$

For the Y -equation: $Y'' - \mu_n^2 Y = 0$ (note the *positive* sign — we are dealing with *hyperbolic*, not oscillatory, behaviour in y). The general solution is

$$Y = A_n \cosh(\mu_n y) + B_n \sinh(\mu_n y).$$

From $Y(0) = 0$: $A_n = 0$. Hence

$$Y_n(y) = B_n \sinh\left(\frac{n\pi y}{a}\right).$$

10.4.2 The Full Solution

Superposing the product solutions $u_n = B_n \sin(n\pi x/a) \sinh(n\pi y/a)$:

$$u(x, y) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right). \quad (16)$$

Applying the remaining boundary condition $u(x, b) = f_2(x)$:

$$f_2(x) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{n\pi b}{a}\right) \sin\left(\frac{n\pi x}{a}\right).$$

This is a Fourier sine series on $[0, a]$ with coefficients $B_n \sinh(n\pi b/a)$. Therefore

$$B_n = \frac{2}{a \sinh\left(\frac{n\pi b}{a}\right)} \int_0^a f_2(x) \sin\left(\frac{n\pi x}{a}\right) dx, \quad n = 1, 2, \dots \quad (17)$$

10.5 General Dirichlet Problem: Superposition of Four Sub-Problems

Now suppose the temperature is prescribed on *all four sides* of the rectangle $[0, a] \times [0, b]$:

$$\begin{aligned} u(x, 0) &= f_1(x), & u(x, b) &= f_2(x), & 0 < x < a, \\ u(0, y) &= g_1(y), & u(a, y) &= g_2(y), & 0 < y < b. \end{aligned}$$

The trick is to **decompose** this problem into four simpler sub-problems, each with non-zero boundary data on exactly one side and zero on the other three:

- Sub-problem #1: $u_{xx} + u_{yy} = 0$, $u(x, 0) = f_1(x)$, others zero.
- Sub-problem #2: $u_{xx} + u_{yy} = 0$, $u(x, b) = f_2(x)$, others zero.
- Sub-problem #3: $u_{xx} + u_{yy} = 0$, $u(0, y) = g_1(y)$, others zero.
- Sub-problem #4: $u_{xx} + u_{yy} = 0$, $u(a, y) = g_2(y)$, others zero.

By linearity, the solution of the original problem is $u = u_1 + u_2 + u_3 + u_4$.

Each sub-problem is solved by the same method as above (separation of variables), with appropriate adjustments of orientation. The four solutions are:

$$\begin{aligned}
u_2(x, y) &= \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right), \quad B_n = \frac{2}{a \sinh\left(\frac{n\pi b}{a}\right)} \int_0^a f_2(x) \sin\left(\frac{n\pi x}{a}\right) dx, \\
u_1(x, y) &= \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi(b-y)}{a}\right), \quad A_n = \frac{2}{a \sinh\left(\frac{n\pi b}{a}\right)} \int_0^a f_1(x) \sin\left(\frac{n\pi x}{a}\right) dx, \\
u_4(x, y) &= \sum_{n=1}^{\infty} D_n \sin\left(\frac{n\pi y}{b}\right) \sinh\left(\frac{n\pi x}{b}\right), \quad D_n = \frac{2}{b \sinh\left(\frac{n\pi a}{b}\right)} \int_0^b g_2(y) \sin\left(\frac{n\pi y}{b}\right) dy, \\
u_3(x, y) &= \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi y}{b}\right) \sinh\left(\frac{n\pi(a-x)}{b}\right), \quad C_n = \frac{2}{b \sinh\left(\frac{n\pi a}{b}\right)} \int_0^b g_1(y) \sin\left(\frac{n\pi y}{b}\right) dy.
\end{aligned}$$

Theorem 10.1 (Solution of the General Dirichlet Problem in a Rectangle). *The solution of the Dirichlet problem for Laplace's equation on $[0, a] \times [0, b]$ with arbitrary boundary data f_1, f_2, g_1, g_2 is*

$$u(x, y) = \sum_{n=1}^{\infty} \left[A_n \sin \frac{n\pi x}{a} \sinh \frac{n\pi(b-y)}{a} + B_n \sin \frac{n\pi x}{a} \sinh \frac{n\pi y}{a} + C_n \sin \frac{n\pi y}{b} \sinh \frac{n\pi(a-x)}{b} + D_n \sin \frac{n\pi y}{b} \sinh \frac{n\pi x}{b} \right], \quad (18)$$

with coefficients A_n, B_n, C_n, D_n given by the Fourier sine integrals above.

Remark 10.1. The key difference between this and the heat/wave problems is the appearance of $\sinh$ (hyperbolic sine) rather than $e^{-\lambda^2 t}$ or $\cos(\lambda ct)$. This reflects the *elliptic* character of Laplace's equation: there is no time variable, and the “transverse” direction in separation of variables gives hyperbolic functions, not decaying or oscillatory ones.

Remark 10.2. The solution (18) is the steady-state temperature in a rectangular plate whose four edges are maintained at prescribed temperatures. Physically, heat flows from hotter regions to cooler ones, and the steady state is the unique harmonic function matching the given boundary values.